Hsuan-Fu (Jeff) Hsu

AI and ML Engineering | Retrieval, RAG and Model Evaluation

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science — Jan 2026 - May 2027 (Expected)

National Taipei University — New Taipei City, Taiwan
Bachelor of Science in Computer Science — Sept 2020 - June 2024

SKILLS

Proficient: Python, C++, C#, Git

Familiar: PyTorch, Sentence Transformers, FAISS, FastAPI, NumPy, Docker, pytest, GitHub Actions, AWS, SQL, TypeScript

Exposure: TensorFlow, tsfresh, Ollama, OpenAI Agents SDK, LangChain, LangGraph, Google Cloud Run, Vitest

EXPERIENCE

National Taipei University — New Taipei City, Taiwan
Undergraduate Researcher | Advisor: Prof. Bor-Shing Lin — July 2022 - Feb 2024

- Built a **Python** pipeline to process NIR signals from 30 participants and compared five ML and four DL models. The team's best model estimated uric acid levels with 10.3% MAPE in leave-one-subject-out testing.
- Received an individual NSTC research grant and placed fourth in NTPU's project competition as part of a three person team.

Tech-Management Workshop Co., Ltd. — Taichung, Taiwan
Software Engineer Intern — July 2026 - Aug 2026

- Built a birth grant application system using **ASP.NET Core MVC** and **SQL Server** as the sole developer, and deployed it to 29 government offices.
- Optimized SQL Server processing and storage, reducing database latency by at least 50% and resource usage by at least 30%.

PROJECTS

Scientific Claim QA: RAG Evaluation — Oct 2026
Python, FastAPI, FAISS, Ollama, Docker | GitHub

- Built a **FastAPI** RAG service with FAISS and a local LLM through Ollama, and tested six setups on 300 claims. A reranker raised nDCG@10 from 0.645 to 0.695, but faithful answers stayed between 79% and 84% in all six.
- Dropped an LLM judge after it matched 20 hand labels no better than chance (kappa −0.07). Shipped the service with API keys, rate limits, **Docker**, and 112 tests in CI.

Scientific Retrieval Fine-tuning — Personal Fork — Oct 2026
Python, PyTorch, Sentence Transformers | GitHub

- Fine tuned **MiniLM** with **PyTorch** on mined hard negatives, improving nDCG@10 by 3.2% and Recall@10 by 2.1% over random negatives on 300 SciFact test claims across three seeds.
- Added data leakage checks, a false negative review, and 53 unit tests.

Promptfoo Agent Retry Validation — Personal Fork — Oct 2026
TypeScript, Vitest, OpenAI Agents SDK | GitHub

- Added nested error paths to the retry policy checks in **TypeScript**. Six new tests fail before the change and pass after it.
- Wrote 81 **Vitest** cases, including 60 config checks and 11 retry scenarios with mocked model calls.

Classroom AC Setpoint Simulator — USC IEEE HackIoT (team of 4) — Oct 2026
TypeScript, Next.js, Raspberry Pi 4, YOLO | GitHub

- Ported a Python air-conditioning model to **TypeScript** and verified matching results across eight weeks of simulated operation. Built a Next.js simulator that estimated 27% energy savings for one room over that period.
- Set up a **Raspberry Pi 4** with teammates to count classroom occupants using YOLO at approximately 7 FPS and log DHT11 temperature and humidity readings to Supabase every 30 seconds.

PUBLICATION

Coauthor, "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, 2026, doi: 10.1109/THMS.2026.3724413.